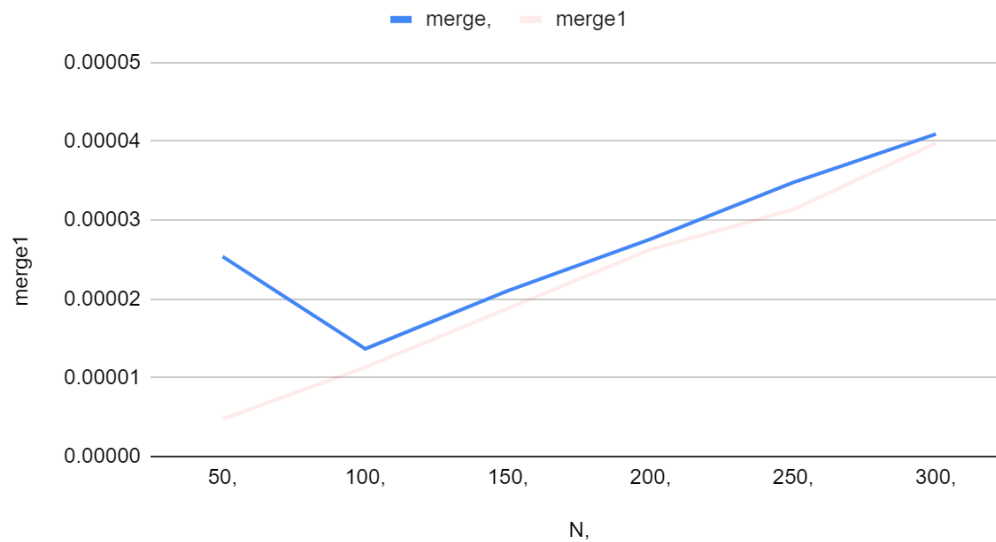


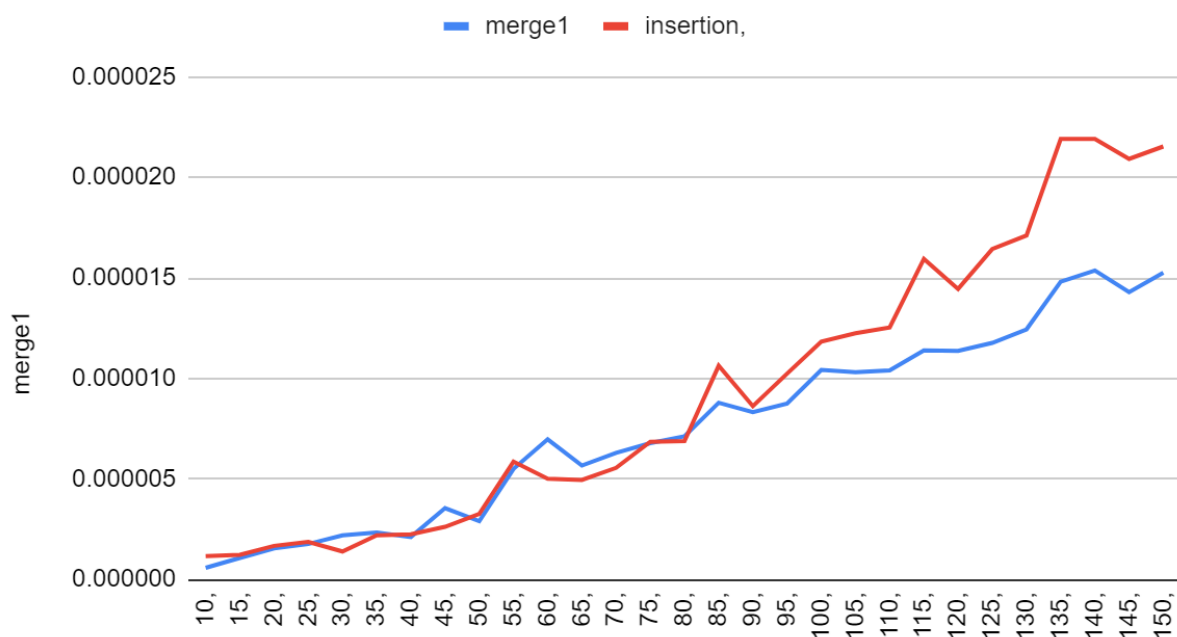
My merge1 out performed normal merge sort at every count of N. The difference however is relatively insignificant because it only changes the number of assignments by roughly high.

merge1 vs. merge,



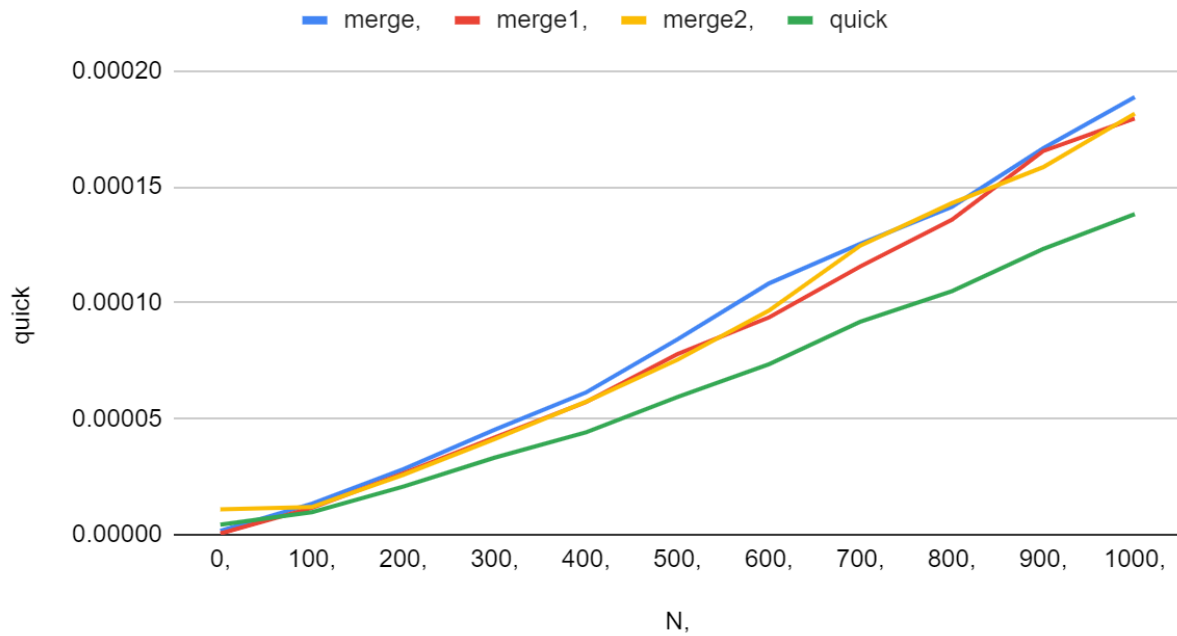
I choose my threshold at 15 because it was the closest point that I could see after which most of the points for merge one were below the insertion sort with a few points that went back over even though merge one started lower than insertion the next points were much closer so with more test I found that it was a rarer case.

merge1 vs. insertion,

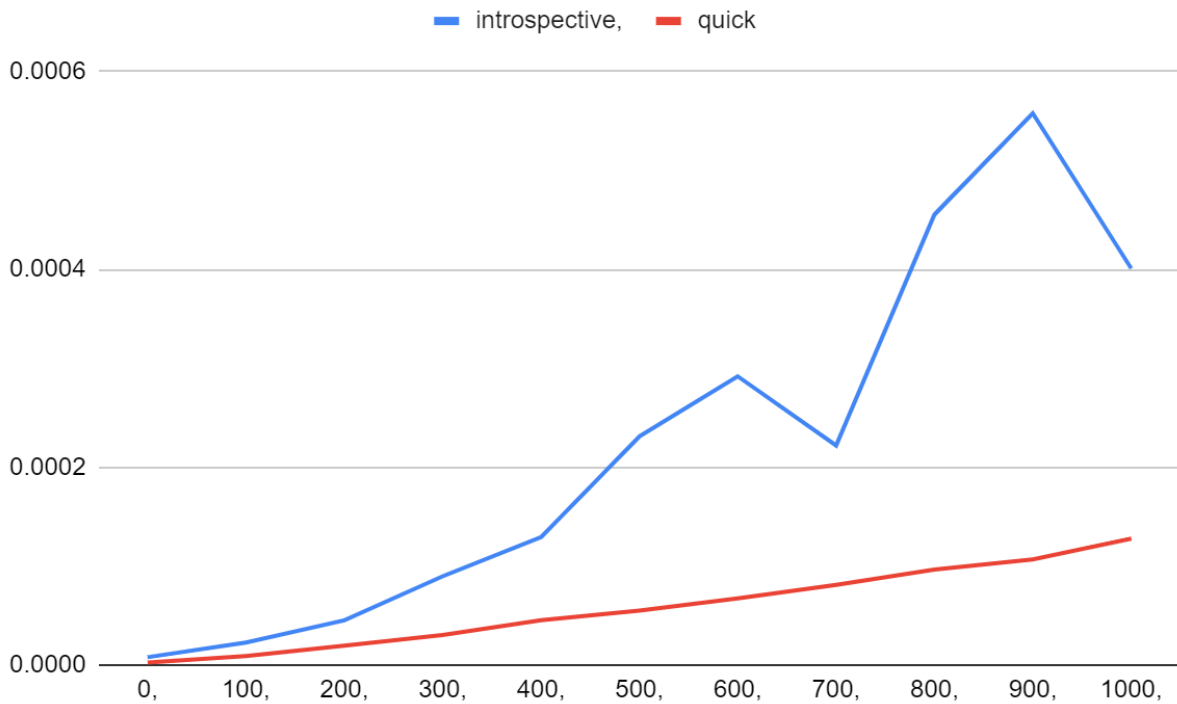


Quick sort here was the fastest at almost all points so it likely didn't have any pathological inputs. While all of the merges performed similarly with each doing better at various points

quick vs. N,



Introspective Sort was much slower than quick for every run im not sure why this happened.



For all the pathological runs introspective was faster with quick appearing to be n^2

